Neel Gali

Software Quality Assurance Engineer

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Professional Experience

Senior Software Engineer *Jul 2025 – Present*

Epik Solutions | Electrification Test Automation Platform

- Designed and implemented comprehensive **Python** test automation frameworks for **battery management** systems, integrating **dSPACE ControlDesk** for **HIL** validation, reducing manual test effort by 40%.
- Led **Agile** scrum teams in applying **Behavior Driven Development** (**BDD**) to define and execute test cases for **electric motor** control software, accelerating feature delivery by 25%.
- Developed sophisticated **C/C++** software for **embedded controllers** validating **inverter** performance in non-automotive applications, ensuring adherence to safety and functional requirements 99.9% of the time.
- Engineered **continuous automation** pipelines using **Jenkins** for **Software-in-the-Loop** (**SIL**) testing across 50+ diverse test environments, improving code release frequency by 30%.
- Analyzed complex software defects in **Vehicle Mechatronic Embedded Controls** (**VMEC**), determining root cause and implementing robust solutions, decreasing critical bug reoccurrence by 35%.
- Mentored junior engineers on **Test Driven Development** (**TDD**) principles and advanced **Python** scripting for **Hardware-in-the-Loop** (**HIL**) testing, elevating team's quality assurance capabilities 20%.
- Collaborated with cross-functional teams to define effective acceptance criteria for new **Electrification Solutions New Markets** (**ESNM**) products, ensuring 95% test coverage.
- Authored detailed test plans and executed them on various test environments, utilizing **MATLAB** for simulation and data analysis, identifying performance issues 20% earlier.

Software Engineer *Jul 2022 – Jun 2025*

Samay Systems LLC | Embedded Systems Verification

- Developed and maintained **Python** scripts for automated **regression testing** of **embedded software** across 3 product lines, reducing testing cycles from 8 hours to 2 hours.
- Performed in-depth debugging and **root cause analysis** on **C/C++** firmware for **battery management** systems, resolving 150+ critical software anomalies over three years.
- Implemented **Hardware-in-the-Loop** (**HIL**) test setups for validating **electric motor** control algorithms, ensuring product reliability above industry standards for 98% of tests.
- Contributed to the design and execution of comprehensive test plans for **inverter** control units, identifying missed requirements and improving system stability by 15%.
- Integrated test suites into **CI/CD** pipelines using **Jenkins** to automate build verification, reducing manual intervention in testing processes by 60%.
- Collaborated with product owners and developers following **Agile** methodologies to ensure high-quality software delivery for embedded applications, improving team velocity by 10%.
- Diagnosed and resolved complex **Software-in-the-Loop** (**SIL**) simulation issues for various **embedded controllers**, enhancing simulation accuracy by 20% for future development.
- Developed tools using **Python** and **VBA** for parsing test logs and generating detailed reports, improving defect tracking and communication efficiency by 25%.
- Ensured compliance with **functional** and **safety requirements** by rigorously testing software components, contributing to successful certification for 5 new product releases.

Software Engineer *Jul 2020 – Aug 2022*

Zemoso Technologies | Software Quality Assurance

- Executed manual and automated test cases for enterprise applications, validating new features and bug fixes across diverse platforms, improving software quality by 15%.
- Developed basic automation scripts using **Python** for repetitive testing tasks, streamlining the QA process and reducing test execution time by 20%.
- Performed thorough black-box and white-box testing on **Java**-based microservices, identifying critical defects prior to production deployment, preventing 10+ major incidents.
- Collaborated with development teams to analyze software defects, document **root causes**, and verify resolutions, contributing to a 10% reduction in post-release issues.
- Authored and maintained detailed test documentation including test plans and test reports, improving clarity and traceability for future quality assurance efforts by 30%.
- Participated actively in sprint planning and daily stand-ups, providing valuable QA insights following **Agile** principles, ensuring alignment with project requirements 95% of the time.

Skills

Languages: Python, C/C++, Java, VBA

Frameworks & Tools: dSPACE ControlDesk, Jenkins, MATLAB

Concepts: Agile, Behavior Driven Development (BDD), Test Driven Development (TDD), Hardware-in-the-Loop (HIL), Software-in-the-Loop (SIL), CI/CD, Root Cause Analysis, Regression Testing, Functional Requirements, Safety Requirements, Continuous Automation

Education

Master's in Computer Science, Oklahoma State University, Stillwater
Bachelor's in Computer Science, JNTU Hyderabad, India